CT-KAE: the two candidate models

Reference and back001, mixture readout, ablations, data assimilation and post-processing

Prepared for R. Grozavescu

28 October 2026

The short version

Two candidates. Both are the same continuous-time Koopman autoencoder (CT-KAE) with exactly linear latent dynamics, read out with a train-calibrated *mixture* readout:

- the **reference**: trained on reconstruction, 8-lead prediction, latent consistency and amplitude matching;
- **back001**: the same, plus the original KAE's *backward latent consistency* at weight 0.01.

(MSE $\times 10^{-3}$, all frames; DA: mean over 37 df9 points; lower is better)	Tra_ext	Tra_int	Tra_long	DA df9
ACDM (official)	2.14	2.92	22.63	0.258
U-Net$_{ut}$ (official)	1.60	**1.48**	22.18	0.270
U-Net$_{tn}$ (official)	**1.37**	1.77	22.38	–
Original KAE (final checkpoint)	2.05	11.05	18.47	0.103
CT-KAE reference + mixture (mean of 3 seeds)	1.77	3.16	13.03	0.095
back001 + mixture (mean of 3 seeds)	1.88	2.89	13.45	**0.088**
With post-processing (§6; forecasts only, train-calibrated)				
reference, 3-seed average + mixture + shrink	1.58	2.92	**11.97**	–
back001, 3-seed average + mixture + shrink	1.69	2.66	12.04	–

- **Long rollouts:** both candidates beat every baseline by about 40%.
- **Data assimilation:** both are about 3× more accurate than ACDM and U-Net, and back001 is the best model tested, better than the reference on 36 of 37 df9 points.
- **Short-horizon forecasts (ext, int):** the U-Nets remain best; back001 + mixture is the best CT-KAE on int and just ahead of ACDM there.
- **back001 against the reference** (paired, 3 seeds): DA -0.0076 ± 0.0040, int -0.27 ± 0.10, ext $+0.11 \pm 0.20$ (tie), long $+0.42 \pm 0.12$. It improves the two metrics ranked first (DA, then MSE on int) and costs 3% on long; it fails the pre-registered rule (lower DA *and* no worse on ext/int/long) on long only.
- **LSiM** gets worse with the mixture readout for both models (it softens long-lead fields); on LSiM the baselines win throughout.
- **Post-processing** (§6): averaging the three seeds is the largest gain (ext −10%, int −8%, long −3 to −4%), and shrinking the long-lead forecast towards a Mach-matched training climatology adds a further 6–9% on long. Together they take the reference to 1.58 / 2.92 / 11.97 and back001 to 1.69 / 2.66 / 12.04. Both cost LSiM.

Recommendation. Keep both. Use **back001 + mixture** where data assimilation and short/medium-range accuracy matter (it is the DA model of the paper), and the reference + mixture as the best long-rollout model. For the best forecasting numbers, report each as a **3-seed average** with the mixture readout and **climatology shrinkage**: every member stays an exact linear model, everything is calibrated on training data, and the cost is 3× inference. Report the single-model numbers alongside, and LSiM for both, since the post-processing lowers MSE by smoothing. With shrinkage, back001's long-rollout deficit against the reference drops from 0.42 to 0.09 (single-seed means).

Contents

1 The model **2**
- 1.1 Architecture . . . 2
- 1.2 The latent dynamics are exactly linear . . . 2
- 1.3 Training . . . 3
- 1.4 Mixture readout (inference only) . . . 3
- 1.5 Data assimilation . . . 3

2 Comparison with the baselines **3**

3 back001 against the reference **3**
- 3.1 Forecasting, lead by lead . . . 3
- 3.2 Data assimilation . . . 5
- 3.3 Zero-shot time resolution and Mach sensitivity . . . 5

4 The mixture readout **6**

5 Ablation study **6**

6 Post-processing study **9**
- 6.1 Results . . . 9

7 Findings from the investigation **11**

8 Open items **11**

1 The model

1.1 Architecture

$$z_0 = E(x_{t-1}, x_t; \mathrm{Ma}), \qquad z(\tau) = e^{\tau A(\mathrm{Ma})} z_0, \qquad \hat{x}(t+\tau) = D(z(\tau); \mathrm{Ma}).$$

- **Encoder** (12.1 M parameters). Input: 2 frames × 4 fields, the obstacle mask and two coordinate channels (11 channels, 64×128). Three CNN stages (widths 64/128/256), two residual blocks each (GroupNorm, SiLU, 3×3 convolutions), stride-2 downsampling to 8×16, flatten, one linear layer to $z \in \mathbb{R}^{256}$. Every residual block is FiLM-modulated by a 64-dimensional embedding of $(\mathrm{Ma} - 0.7)/0.1$, zero-initialised.
- **Latent dynamics** (21 k parameters). 128 independent damped oscillators, $A = \mathrm{blockdiag}_j \left(\begin{smallmatrix} -\gamma_j & -\omega_j \\ \omega_j & -\gamma_j \end{smallmatrix} \right)$, $\gamma_j(\mathrm{Ma}) = \mathrm{softplus}(\cdot) \geq 0$ (no growing modes), $\omega_j(\mathrm{Ma})$ free; both are a base value plus a zero-initialised Mach correction. No fixed-point offset.
- **Decoder** (12.6 M parameters). Mirror of the encoder with FiLM; the only nonlinearity between z and the pixels.

1.2 The latent dynamics are exactly linear

$e^{\tau A}$ is evaluated in closed form, $e^{-\gamma_j \tau}$ times a rotation by $\omega_j \tau$ per block; this *is* the matrix exponential. Checked on the trained back001 (seed 0, Ma 0.66) against `torch.linalg.matrix_exp` in float64:

τ (time units)	0.1	1.0	5.8	23.8	−5.0	−23.8
rel. difference, float64	2e-16	1e-15	5e-15	2e-14	6e-15	3e-14
rel. error, float32 (as used)	5e-8	6e-8	4e-7	1e-6	3e-7	4e-6

Semigroup $e^{1.0A} e^{5.8A} = e^{6.8A}$: 6×10^{-16}; linearity: 3×10^{-17}; largest real part of the spectrum: -4.5×10^{-5}. The block form loses no generality beyond growing modes, Jordan blocks and paired equal real eigenvalues (any diagonalisable real A is similar to it, and the similarity is absorbed by the encoder and decoder).

1.3 Training

Term	Weight	What it enforces
rec	1	$D(E(\text{window}))$ reproduces the window's last frame
pred	1	$D(z(\tau_k))$ matches the true frame at each of the 8 leads (0.1 apart)
lat	0.1	$z(\tau_k)$ matches the stop-gradient encoding of the true window at τ_k (normalised by $E\|z\|^2$)
amp	0.1	oscillator amplitudes $\|(a_j, b_j)\|$ match the encoded ones
back (back001 only)	**0.01**	encode the last window of the clip, propagate *backward* to every earlier lead and to t_0, match the stop-gradient encodings

Channel-weighted squared error; 20 k steps, batch 16, AdamW (3×10^{-4}, weight decay 10^{-4}, 500-step warm-up, cosine to 5%), gradient clip 1, bf16. Data: the 33 ACDM training simulations (train.nc plus simulation 40), all frames, random offsets, no validation split, final checkpoint (the ACDM protocol). 24.7 M parameters; about 3 GPU hours per seed.

1.4 Mixture readout (inference only)

The forecast averages 5 decodes of the same linear flow run with all frequencies scaled by $s_q = 1 + \sigma\xi_q$ (Gauss–Hermite nodes and weights): $\hat{x}(\tau) = \sum_q w_q\, D\big(e^{\tau A_{s_q}} z_0\big)$. Every decoded latent is a real trajectory of the model; the averaging is in pixel space. σ is chosen per checkpoint on training rollouts only (§4). Training is unchanged, the same checkpoint gives both readouts ($\sigma = 0$ is the single decode), and DA does not use the readout.

1.5 Data assimilation

Strong-constraint latent 4D-Var: the control is z_0 only, the misfit $\|HD(e^{\tau_j A} z_0) - y_j\|^2$ over observations 0.9–5 time units after t_0 (df9) or from one frame after t_0 (df1) is minimised with Adam from $z = 0$ for 500 iterations, learning rate tuned per model on gt_extrap; the analysis $D(z_0)$ at the unobserved t_0 is scored on gt_interp (8 problems per point).

2 Comparison with the baselines

Table 1: Test set, same cases and scorers for every model. MSE $\times 10^{-3}$ on all frames (ACDM-paper convention); LSiM $\times 10^{-1}$; DA: mean analysis error over the 37 points of the six df9 sweeps (noise, sparsity, window length, number of observation times, misspecified noise, joint sparsity). CT-KAE rows: mean ± s.d. over 3 seeds. Lower is better. *Selected on forecast loss on gt_interp, the trajectories DA is tested on; its DA is therefore optimistic. Original KAE forecasts: final checkpoint re-run through the verified pipeline (the stored prediction files are not reproducible); its LSiM was not computed.

	MSE			LSiM			DA
Model	Tra_ext	Tra_int	Tra_long	Tra_ext	Tra_int	Tra_long	df9 (37 pts)
ACDM (official, r20)	2.14	2.92	22.63	1.29	1.38	3.83	0.258
U-Net$_{ut}$ (official, unrolled 8)	1.60	1.48	22.18	1.09	0.95	3.81	0.270
U-Net$_{tn}$ (official, train noise 0.01)	1.37	1.77	22.38	1.05	1.05	3.92	–
Original KAE, final checkpoint (ep. 199)	2.05	11.05	18.47	–	–	–	0.103
Original KAE, selected checkpoint (ep. 40)*	–	–	–	–	–	–	0.090
CT-KAE reference, single decode	1.91±0.30	3.37±0.14	16.08±0.59	1.41±0.03	1.71±0.04	4.06±0.05	0.095±0.003
CT-KAE reference + mixture	1.77±0.23	3.16±0.09	13.03±0.19	1.72±0.09	1.87±0.07	5.74±0.12	0.095±0.003
back001, single decode	2.03±0.21	3.04±0.14	15.11±0.99	1.49±0.08	1.71±0.04	4.88±0.38	0.088±0.006
back001 + mixture	1.88±0.19	2.89±0.11	13.45±0.26	1.81±0.04	1.83±0.04	6.09±0.43	0.088±0.006

3 back001 against the reference

3.1 Forecasting, lead by lead

- **Leads 1–15: better everywhere.** Lead-1 MSE 0.122→0.100 (ext, −18%), 0.259→0.238 (int, −8%), 0.382→0.327 (long, −15%), negative in nearly every seed. Lead 1 is almost pure encoding and decoding, so the backward term mainly yields a more accurate encoder.
- **int, leads 31–58: better**, -0.51 ± 0.23 (anomaly correlation 0.804→0.823).

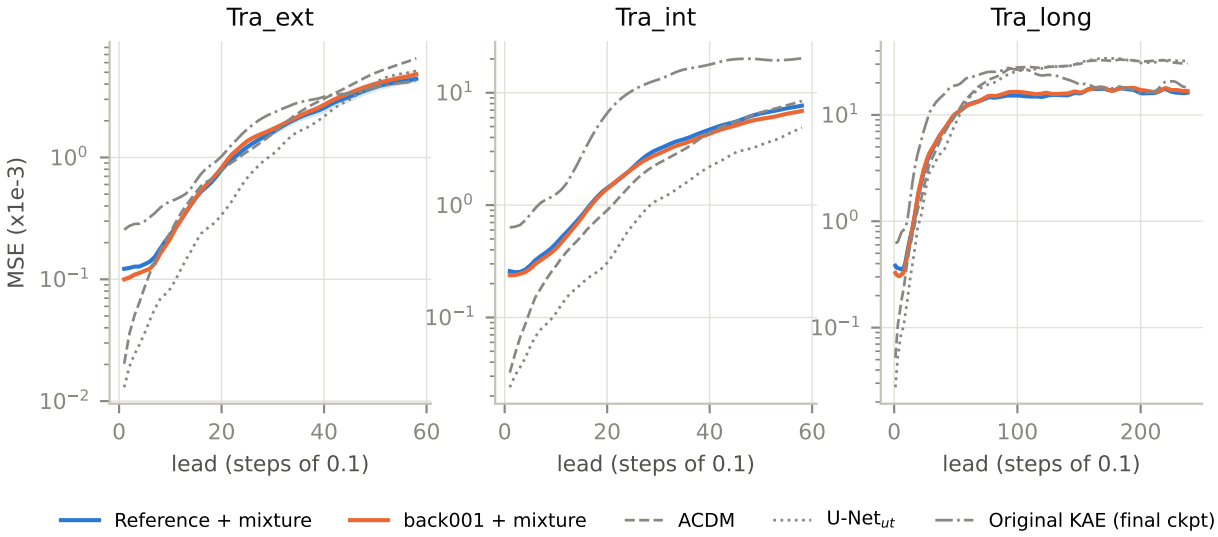


Figure 1: MSE per lead (log scale), CT-KAE: mean over 3 seeds with ±SE band, both with the mixture readout. The CT-KAEs start above the U-Nets at short leads (the encoding floor, §7) and stay far below every baseline beyond about 60 leads.

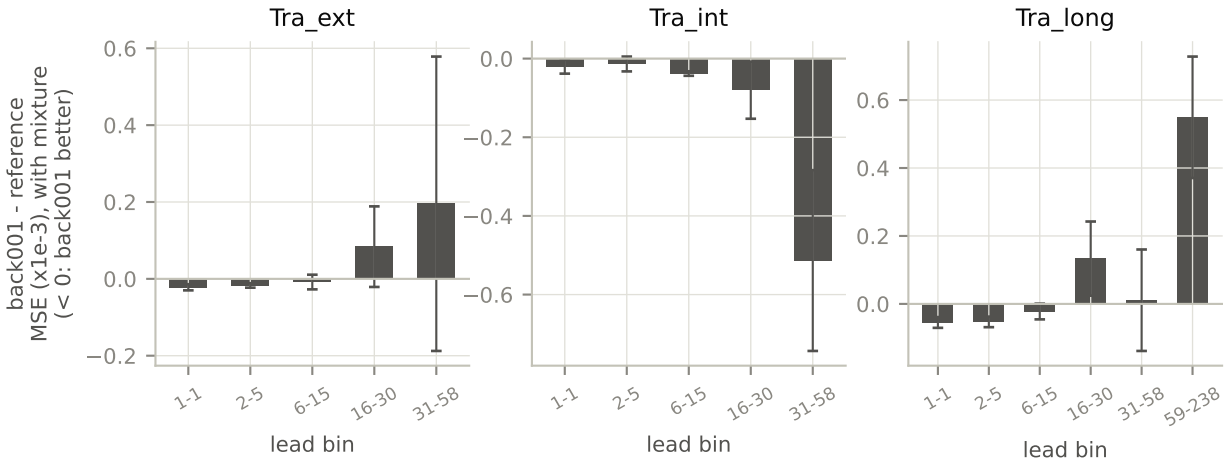


Figure 2: Paired difference back001 − reference by lead bin, mixture readout, mean ± SE over 3 seeds; negative means back001 is better.

- **ext, leads 16–58: worse on average** (+0.08, +0.20) but one seed reverses sign: seed noise rather than a consistent loss.
- **long, leads 59–238: the only real loss**, $+0.55 \pm 0.18$ with the mixture (all three seeds), although without the mixture back001 is better there (-1.29 ± 1.19, 19.69→18.40). back001 gains less from the mixture (-2.05 against -3.9).

Why long gets worse. back001 learns more damping (median γ 0.058 against 0.045, e-folding time 17 against 22 time units; $|\omega|$ range almost unchanged). Part of its long-lead hedging is therefore latent damping, the weaker of the two hedges (pixel-space averaging of phases beats latent shrinkage, §7), and the mixture has less phase error left to average. The same pattern appears on 238-lead *training* rollouts (single decode 11.07 vs 11.05, best mixture 9.93 vs 10.11; Figure 4), so it is a property of what back001 learned, not of the readout calibration: re-calibrating σ on 238-lead training rollouts returns the same σ (0.04 / 0.04 / 0.05) for all six checkpoints.

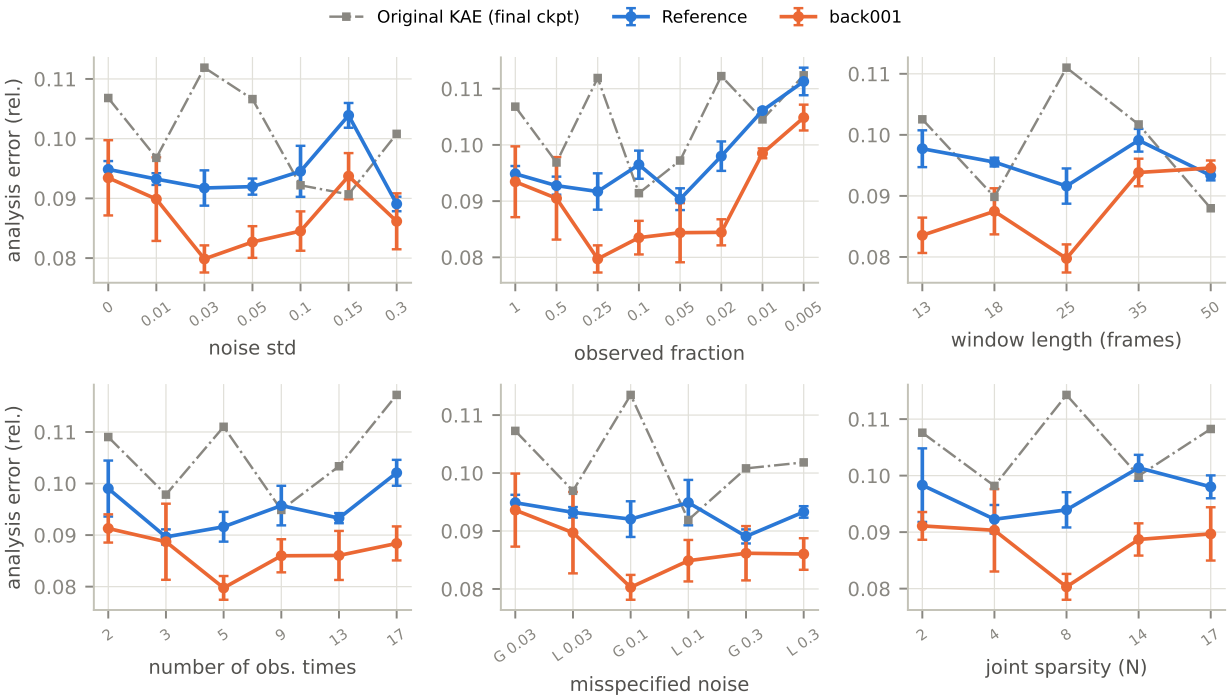


Figure 3: DA analysis error over the six df9 sweeps, mean ± SE over 3 seeds; the original KAE is its final checkpoint on the same DA code. ACDM and U-Net (0.26–0.27) are off scale.

3.2 Data assimilation

df9 sweep (3 seeds, paired)	points	reference	back001	Δ±SE	back001 better
noise	7	0.0942	0.0872	-0.0070 ± 0.0044	7/7
sparsity	8	0.0977	0.0899	-0.0078 ± 0.0035	8/8
window length	5	0.0955	0.0878	-0.0076 ± 0.0027	4/5
number of observation times	6	0.0952	0.0867	-0.0085 ± 0.0047	6/6
misspecified noise	6	0.0929	0.0868	-0.0061 ± 0.0042	6/6
joint sparsity	5	0.0968	0.0880	-0.0088 ± 0.0048	5/5
all	37	0.0954	**0.0878**	$\mathbf{-0.0076 \pm 0.0040}$	**36/37**
df1 headline (seed 0)	1	0.0286	0.0254	−11%	–
df1 sweeps (seed 0)	47	0.041–0.047	0.030–0.035	about −25%	46/47

- The gain is largest with noisy (0.1), heavy-tailed (Laplace) and sparse observations and short windows, and vanishes for the longest window (50 frames), where long-range forecast error dominates the misfit.
- Seed caveat: seed 0 gains least (−0.0001 to −0.003); seeds 1–2 carry most of the improvement. df1 is seed 0 only, as for the reference.
- **Not from observability.** The DA conditioning diagnostic shows unchanged observation gains (median 3.06 vs 2.98; 10th percentile 0.44 vs 0.44), a latent problem about 3× *worse* conditioned for the optimiser (condition number 6.7×10^4 vs 2.2×10^4), and a lower misfit at the analysis (0.0016 vs 0.0020, with the misfit of the true state 0.0024 for both: 4D-Var still prefers a wrong state). The gain comes from what the latent encodes, consistent with the lead-1 gain.

3.3 Zero-shot time resolution and Mach sensitivity

Trained on frames 0.1 apart, queried every 0.05 (3 seeds, MSE $\times 10^{-3}$):

	int, half-step leads	int, whole-step leads	ext, half-step	ext, whole-step
reference	3.51	3.49	1.99	1.98
back001	3.16	3.15	2.11	2.10
Δ±SE	-0.35 ± 0.14	-0.35 ± 0.14	$+0.12 \pm 0.25$	$+0.12 \pm 0.25$

Both models are as accurate at never-trained half steps as at trained steps. Output change under a wrong Mach number (true 0.66) is nearly identical; back001 reacts slightly more at 0.90 (rms 0.76 vs 0.68).

4 The mixture readout

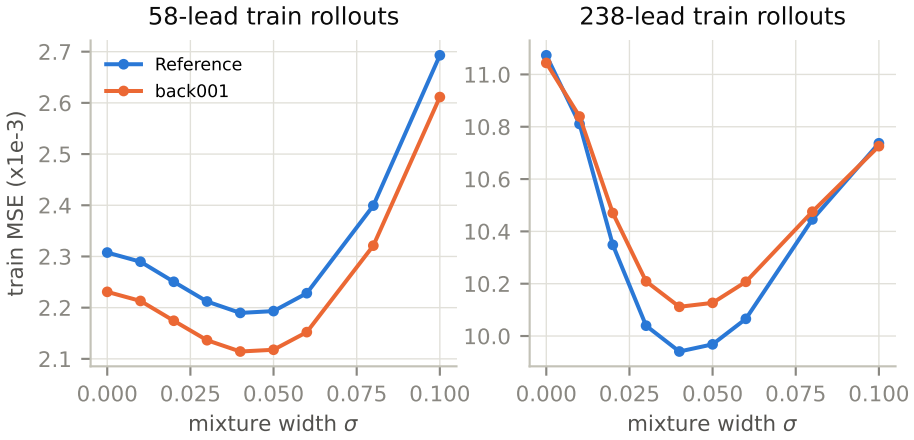


Figure 4: Calibration of the mixture width on training rollouts only (mean of 3 seeds).

- **Why.** The wake is a noisy oscillator: its phase speed varies by about ±14% within a simulation and cannot be predicted from the input window ($R^2 \approx 0.19$ even from the full state). A single decode commits to one phase and is wrong at long leads.
- **Calibration.** $\sigma \in \{0, \ldots, 0.10\}$ on 58-lead rollouts from two windows per training simulation; selected 0.04 / 0.04 / 0.05 for the reference seeds and the same for back001. No test data are used.
- **Effect on the reference** (3 seeds, paired): ext -0.14 ± 0.04, int -0.22 ± 0.03, long -3.04 ± 0.31; DA unchanged.
- **Cost.** 5× the decoding at inference, and lower energy at long leads (about 0.5–0.6 of the true fluctuation energy at lead 58), so LSiM gets worse: ext 1.41→1.72, int 1.71→1.87, long 4.06→5.74 ($\times 10^{-1}$).

5 Ablation study

Each variant changes one thing in the reference and is trained with the same seeds; entries are the paired change against the reference run of the same seed (mean over seeds ± SE of the paired change). Bold: improvement beyond 2 SE; †: deterioration beyond 2 SE (2-seed SEs are rough).

Tables 2 and 3 are on landscape pages at the end of this section. DA learning rates were re-tuned wherever the optimum sat at the edge of the grid; every row's optimum is now interior.

Table 2: Ablations, **single decode**. MSE $\times 10^{-3}$ (all frames), LSiM $\times 10^{-1}$, DA = mean over the 26 points of the four df9 sweeps (window length, number of observation times, sparsity, noise).

Variant	seeds	ext MSE	int MSE	long MSE	ext LSiM	int LSiM	long LSiM	DA df9
Reference (absolute)	3	1.91	3.37	16.08	1.41	1.71	4.06	0.096
Architecture and training								
- Mach conditioning (encoder, decoder, generator)	3	+8.89±2.30†	+9.64±1.46†	+4.61±1.83†	+2.18±0.26†	+2.07±0.17†	+2.01±0.29†	+0.096±0.005†
Generator: dense, unconstrained	3	-0.09±0.13	+0.03±0.05	+18.13±1.65†	+0.06±0.03†	+0.06±0.04	+2.63±0.13†	+0.024±0.004†
Generator: Hurwitz (J - R)	3	+0.06±0.17	+0.07±0.06	+0.14±0.86	-0.01±0.03	-0.02±0.01	+0.04±0.19	+0.041±0.001†
Latent 128	3	+0.04±0.13	+0.45±0.07†	-0.19±0.79	+0.07±0.07	+0.15±0.01†	+0.38±0.23	+0.013±0.003†
Latent 512	3	-0.18±0.12	+0.11±0.39	+1.71±0.99	-0.06±0.05	+0.09±0.15	+0.22±0.31	+0.006±0.005
1 input frame (no history)	3	+0.77±0.17†	+0.25±0.07†	-0.16±0.24	+0.18±0.04†	+0.07±0.04	+0.13±0.10	+0.003±0.002
One-step training ($2 \to 1$)	3	+1.37±0.16†	+2.73±0.27†	+0.57±0.95	+1.43±0.10†	+0.96±0.16†	+0.95±0.29†	+0.007±0.003†
+ conditional fixed point μ(Ma)	3	+0.22±0.15	**-0.31±0.11**	-0.13±0.24	+0.04±0.03	+0.02±0.04	+0.03±0.21	-0.001±0.001
Loss terms used (removed)								
- reconstruction loss	2	+0.05±0.08	+0.37±0.22	+0.77±0.52	+0.03±0.11	+0.05±0.03	+0.17±0.49	+0.025±0.003†
- prediction loss	2	+2.56±0.22†	+4.33±0.46†	+6.09±1.91†	+0.77±0.08†	+1.05±0.12†	+0.21±0.16	+0.013±0.006†
- forward latent consistency	2	+5.11±0.74†	+10.26±1.80†	+14.19±3.58†	+2.29±0.21†	+1.90±0.36†	+1.81±0.31†	+0.049±0.002†
- amplitude loss	2	+0.24±0.19	-0.01±0.35	-0.10±1.10	+0.02±0.03	+0.04±0.04	+0.10±0.32	+0.005±0.000†
Loss terms of the original KAE (added)								
+ backward latent consistency (0.1)	2	+1.14±0.23†	+0.77±0.07†	+1.61±1.06	+0.57±0.02†	+0.25±0.05†	+0.33±0.08†	**-0.018±0.003**
+ backward latent consistency (0.03)	3	+0.31±0.07†	-0.02±0.07	+1.07±0.89	+0.12±0.04†	+0.02±0.04	+0.33±0.06†	**-0.013±0.003**
+ backward latent consistency (0.01)	3	+0.11±0.24	**-0.33±0.13**	-0.97±0.88	+0.08±0.05	-0.00±0.02	+0.82±0.22†	**-0.008±0.004**
+ directional cosine (0.1)	2	+0.11±0.04†	+0.12±0.14	-0.51±0.67	-0.01±0.05	+0.00±0.05	+0.19±0.01†	+0.007±0.005
+ latent energy (0.1)	2	+0.24±0.27	-0.07±0.10	-0.50±0.42	+0.03±0.09	+0.01±0.01	+0.12±0.09	-0.001±0.001
+ temporal Sobolev (0.1)	2	+0.05±0.07	-0.13±0.12	**-0.70±0.02**	-0.01±0.04	**-0.03±0.00**	+0.25±0.20	+0.002±0.003
+ spatial Sobolev (0.1)	2	+0.06±0.24	**-0.19±0.00**	**-0.36±0.15**	+0.01±0.04	+0.02±0.00†	+0.09±0.06	+0.004±0.000†
+ spectral loss (0.1)	2	+0.04±0.01†	+0.11±0.06	-0.20±0.49	-0.01±0.01	+0.10±0.01†	+0.20±0.11	+0.012±0.000†
Prediction weighting: cosine over leads (original KAE)	2	+0.31±0.02†	+0.70±0.52	-0.23±0.99	-0.03±0.04	+0.14±0.13	+0.73±0.02†	-0.001±0.002
Regularisation								
Supervise 4 random of the 8 leads (first and last always)	3	+0.15±0.13	**-0.22±0.07**	-0.47±0.41	+0.01±0.02	-0.05±0.03	+0.09±0.08	+0.005±0.002†
Random target times on the 0.05 grid within the 0.8 horizon	3	+0.02±0.22	-0.09±0.14	-0.63±0.35	+0.02±0.04	+0.01±0.03	+0.41±0.09†	+0.010±0.001†
+ Mach-number noise (s.d. 0.005)	3	+0.03±0.10	-0.07±0.06	**-1.07±0.22**	+0.05±0.03	+0.01±0.05	+0.59±0.10†	+0.006±0.002†
+ input-frame noise (s.d. 0.01, normalised units)	3	+0.07±0.19	-0.10±0.09	-0.38±0.42	+0.03±0.03	+0.02±0.04	+0.20±0.28	+0.006±0.004
+ latent noise per step (0.005 × rms z; original disturb_std)	3	+0.06±0.10	-0.05±0.11	**-0.64±0.29**	+0.01±0.06	-0.00±0.02	+0.25±0.04†	+0.003±0.002†
Regularisation combined								
+ latent noise + Mach noise	3	+0.14±0.12	-0.09±0.20	**-0.93±0.13**	+0.02±0.03	-0.01±0.04	+0.45±0.19†	+0.010±0.002†
+ latent noise + Mach noise + 4 random of 8 leads	3	+0.03±0.14	+0.01±0.17	-1.37±0.83	+0.01±0.02	+0.01±0.01	+0.62±0.11†	+0.009±0.001†
Candidate improvements								
Mach conditioning in the generator only (encoder, decoder Mach-free)	2	+0.20±0.04†	+0.31±0.11†	-0.80±0.75	+0.28±0.05†	+0.04±0.02†	+0.84±0.01†	+0.005±0.000†
L1 pixel distance (rec and pred)	3	+0.63±0.03†	+2.00±0.55†	+1.00±1.24	+0.95±0.29†	+0.45±0.09†	+0.77±0.39	+0.027±0.006†
L1 reconstruction, L2 prediction (original KAE)	3	+0.24±0.17	+0.49±0.19†	+0.16±0.58	+0.42±0.08†	+0.18±0.04†	+0.43±0.18†	+0.013±0.003†

Table 3: Ablations, **mixture readout** (each checkpoint with its own train-calibrated σ). DA is identical to Table 2. The dense and Hurwitz generators are absent: the mixture rescales oscillator frequencies, which only the block generator has.

Variant	seeds	ext MSE	int MSE	long MSE	ext LSiM	int LSiM	long LSiM	DA df9
Reference (absolute)	3	1.77	3.16	13.03	1.72	1.87	5.74	0.096
Architecture and training								
- Mach conditioning (encoder, decoder, generator)	3	+8.04±1.82†	+8.83±1.17†	+6.50±1.57†	+2.21±0.05†	+2.42±0.24†	+0.65±0.33	+0.096±0.005†
Latent 128	3	+0.07±0.10	+0.42±0.05†	+0.52±0.25†	+0.15±0.12	+0.18±0.04†	+0.28±0.05†	+0.013±0.003†
Latent 512	3	-0.13±0.10	+0.13±0.38	+2.27±1.31	+0.13±0.04†	+0.10±0.13	-0.41±0.41	+0.006±0.005
1 input frame (no history)	3	+0.74±0.14†	+0.31±0.05†	+0.23±0.17	+0.13±0.05†	+0.02±0.03	-0.12±0.09	+0.003±0.002
One-step training ($2 \to 1$)	3	+1.62±0.10†	+2.65±0.33†	+2.10±0.37†	+1.61±0.09†	+1.12±0.15†	+0.24±0.17	+0.007±0.003†
+ conditional fixed point μ(Ma)	3	+0.19±0.12	**-0.26±0.07**	+0.36±0.06†	+0.02±0.05	+0.00±0.04	-0.02±0.12	-0.001±0.001
Loss terms used (removed)								
- reconstruction loss	2	+0.05±0.09	+0.27±0.19	+0.89±0.46	+0.16±0.00†	+0.08±0.11	-0.02±0.29	+0.025±0.003†
- prediction loss	2	+2.37±0.08†	+4.00±0.54†	+5.94±0.94†	+1.54±0.20†	+1.43±0.16†	**-0.51±0.16**	+0.013±0.006†
- forward latent consistency	2	+4.98±0.34†	+7.85±1.07†	+12.12±2.53†	+2.89±0.29†	+2.33±0.28†	+0.86±0.39†	+0.049±0.002†
- amplitude loss	2	+0.19±0.19	-0.00±0.28	**-0.05±0.01**	+0.11±0.04†	+0.07±0.08	+0.25±0.04†	+0.005±0.000†
Loss terms of the original KAE (added)								
+ backward latent consistency (0.1)	2	+1.06±0.21†	+0.72±0.09†	+1.30±0.63†	+1.16±0.20†	+0.59±0.12†	+0.33±0.17	**-0.018±0.003**
+ backward latent consistency (0.03)	3	+0.26±0.06†	-0.06±0.04	+0.65±0.16†	+0.38±0.09†	+0.08±0.08	+0.32±0.02†	**-0.013±0.003**
+ backward latent consistency (0.01)	3	+0.11±0.20	**-0.27±0.10**	+0.42±0.12†	+0.08±0.04†	-0.03±0.02	+0.34±0.21	**-0.008±0.004**
+ directional cosine (0.1)	2	+0.12±0.02†	+0.16±0.13	+0.22±0.27	**-0.13±0.05**	-0.07±0.06	**-0.27±0.00**	+0.007±0.005
+ latent energy (0.1)	2	+0.22±0.25	-0.08±0.08	+0.09±0.26	-0.00±0.09	+0.01±0.00†	+0.05±0.05	-0.001±0.001
+ temporal Sobolev (0.1)	2	+0.05±0.07	-0.15±0.12	**-0.06±0.01**	-0.03±0.04	**-0.06±0.01**	+0.08±0.05	+0.002±0.003
+ spatial Sobolev (0.1)	2	+0.06±0.22	**-0.17±0.01**	**-0.16±0.06**	+0.01±0.02	+0.00±0.01	+0.09±0.13	+0.004±0.000†
+ spectral loss (0.1)	2	+0.04±0.01†	+0.06±0.02†	-0.10±0.15	+0.15±0.02†	+0.16±0.00†	+0.38±0.01†	+0.012±0.000†
Prediction weighting: cosine over leads (original KAE)	2	+0.27±0.02†	+0.60±0.39	+0.52±0.50	+0.21±0.03†	+0.32±0.09†	+0.74±0.11†	-0.001±0.002
Regularisation								
Supervise 4 random of the 8 leads (first and last always)	3	+0.14±0.11	**-0.18±0.05**	-0.05±0.18	-0.07±0.05	**-0.09±0.02**	-0.05±0.13	+0.005±0.002†
Random target times on the 0.05 grid within the 0.8 horizon	3	+0.03±0.19	-0.06±0.11	-0.01±0.15	+0.00±0.03	-0.01±0.03	+0.26±0.10†	+0.010±0.001†
+ Mach-number noise (s.d. 0.005)	3	+0.05±0.09	-0.06±0.07	-0.13±0.09	+0.07±0.04	+0.01±0.04	+0.42±0.20†	+0.006±0.002†
+ input-frame noise (s.d. 0.01, normalised units)	3	+0.08±0.17	-0.08±0.09	+0.20±0.26	+0.01±0.08	+0.01±0.10	+0.04±0.38	+0.006±0.004
+ latent noise per step (0.005 × rms z; original disturb_std)	3	+0.05±0.09	-0.04±0.08	+0.03±0.12	+0.01±0.07	-0.00±0.01	+0.18±0.05†	+0.003±0.002†
Regularisation combined								
+ latent noise + Mach noise	3	+0.14±0.10	-0.06±0.16	+0.01±0.05	-0.03±0.05	**-0.05±0.02**	+0.12±0.12	+0.010±0.002†
+ latent noise + Mach noise + 4 random of 8 leads	3	+0.04±0.12	+0.03±0.13	-0.19±0.21	-0.06±0.07	**-0.04±0.02**	+0.33±0.01†	+0.009±0.001†
Candidate improvements								
Mach conditioning in the generator only (encoder, decoder Mach-free)	2	+0.22±0.02†	+0.27±0.07†	+0.39±0.20	+0.44±0.03†	+0.07±0.01†	+0.51±0.16†	+0.005±0.000†
L1 pixel distance (rec and pred)	3	+0.94±0.15†	+1.78±0.40†	+1.89±0.74†	+1.64±0.43†	+0.86±0.20†	+0.67±0.12†	+0.027±0.006†
L1 reconstruction, L2 prediction (original KAE)	3	+0.28±0.12†	+0.44±0.17†	+0.24±0.26	+0.64±0.11†	+0.22±0.04†	+0.56±0.03†	+0.013±0.003†

What the ablations show (single decode unless stated):

- **Mach conditioning is essential** (removing it: ext +8.9, int +9.6, DA +0.096). Conditioning only the generator (encoder and decoder Mach-free) is also worse (int +0.31, DA +0.005).
- **The structured generator matters**: a dense unconstrained generator blows up long rollouts (long +18) and costs DA (+0.024); a Hurwitz generator costs DA (+0.041).
- **Latent size 256 is right**: 128 is worse on int (+0.45) and DA (+0.013); 512 is no better anywhere and worse on long with the mixture (+2.27).
- **Two input frames matter**: one frame costs ext (+0.77) and int (+0.25).
- **A Mach-dependent fixed point** $\mu(\mathrm{Ma})$ helps int (-0.31 ± 0.11; with the mixture -0.26) and leaves DA unchanged (-0.001 ± 0.001), but costs ext and long slightly (with the mixture +0.19, +0.36). Not adopted; a candidate to combine with back001.
- **Multi-step training and latent consistency are essential**; one-step training and removing the latent-consistency or prediction loss are worse on almost everything.
- **Backward latent consistency is the only term that improves DA**, with a trade-off set by its weight: 0.1 gives DA -0.018 ± 0.003 but ext +1.14, int +0.77; 0.01 (back001) keeps most of the DA gain at no cost on ext and a gain on int.
- **Every regulariser and combination made DA worse**; the combinations (latent and Mach noise, with or without 4 of 8 leads) improve long (-0.93, -1.37) but cost DA (+0.010, +0.009).
- **L1 instead of squared error is worse** (3 seeds): L1 for reconstruction and prediction DA $+0.027 \pm 0.006$, int $+2.00 \pm 0.55$; L1 for reconstruction only (the original KAE's setting) DA $+0.013 \pm 0.003$, int $+0.49 \pm 0.19$. L1 does not explain the original KAE's DA.
- **Hedged readout** (a learned forecast decay separate from the state dynamics) was dropped before completion: the mixture readout supersedes it.

6 Post-processing study

Goal: the best possible forecasting numbers from the two candidates without changing training or the linear latent model. Every technique is calibrated on training rollouts only (238 leads from two windows per training simulation) and scored once on test.

Technique	Idea
Lead-dependent mixture width $\sigma(\tau)$	phase uncertainty grows with lead; one width for all leads may be too narrow early or late
Wiener / MOS filter	per lead, channel and radial wavenumber, shrink the unpredictable part of the forecast anomaly towards an anchor; gain = regression of truth on forecast. Anchors: the forecast's own lead-1..58 mean (no extra data) and a Mach-interpolated training climatology
Scalar shrinkage	the same with one gain per lead and channel (checks whether scale dependence matters)
Seed ensembles	mean of the 3 seeds of a model, and of all 6 checkpoints; each member is still an exact linear model
Combinations	the above stacked

Caveats. The model fits training data much better than test data (§7), so train-calibrated corrections are conservative. The training Mach numbers skip 0.64–0.68 (where int and long sit) and stop at 0.53 (ext is 0.50–0.52), so the climatology anchor must be interpolated or extrapolated.

6.1 Results

Lab: 238-lead rollouts from two windows of each training simulation give the calibration statistics for nine mixture widths; the test forecasts of all widths are stored once and combined offline (`scripts/postproc/`). The plain mixture reproduces the stored test scores exactly (reference seed 0: 1.534 / 3.124 / 13.214).

Table 4: Post-processing variants, paired against the mixture readout of the same checkpoint (mean over 3 seeds ± SE). First row: absolute all-frames MSE $\times 10^{-3}$ of the mixture. Bold: improvement beyond 2 SE; †: deterioration beyond 2 SE. "Own mean": the forecast's mean over leads 1–58; "climatology": Mach-interpolated time-mean of the training simulations.

	Reference			back001		
Variant	ext	int	long	ext	int	long
mixture (current protocol)	1.77	3.16	13.03	1.88	2.89	13.45
single decode	+0.14±0.04†	+0.22±0.03†	+3.04±0.31†	+0.15±0.01†	+0.15±0.01†	+1.65±0.45†
mixture, lead-dependent $\sigma(\tau)$	+0.00±0.00	+0.00±0.00	+0.11±0.03†	+0.00±0.00	+0.00±0.00	+0.02±0.02
mixture + scalar shrink to own mean	+0.00±0.00	-0.00±0.00	**-0.26±0.02**	-0.00±0.00	-0.01±0.00	**-0.56±0.02**
mixture + Wiener to own mean	-0.00±0.00	**-0.02±0.00**	**-0.36±0.01**	-0.00±0.00	**-0.02±0.01**	**-0.72±0.01**
mixture + scalar shrink to climatology	**-0.01±0.00**	-0.01±0.01	**-0.90±0.05**	-0.01±0.01	-0.02±0.01	**-1.23±0.05**
mixture + Wiener to climatology	**-0.01±0.00**	**-0.02±0.01**	**-0.88±0.07**	-0.01±0.01	**-0.03±0.01**	**-1.25±0.04**
$\sigma(\tau)$ + scalar shrink to own mean	+0.00±0.00	-0.00±0.00	**-0.20±0.02**	-0.00±0.00	-0.01±0.00	**-0.55±0.02**
$\sigma(\tau)$ + Wiener to own mean	-0.00±0.00	**-0.02±0.00**	**-0.28±0.04**	-0.00±0.00	**-0.02±0.01**	**-0.71±0.02**
$\sigma(\tau)$ + Wiener to climatology	**-0.01±0.00**	**-0.02±0.01**	**-0.80±0.10**	-0.01±0.01	**-0.03±0.01**	**-1.23±0.03**
single decode + Wiener to own mean	+0.03±0.01†	+0.02±0.01†	-0.03±0.12	+0.03±0.00†	+0.00±0.00	**-0.59±0.07**

Table 5: Seed ensembles: the field average of the 3 seeds of a model, or of all 6 checkpoints; each member uses the variant named in the row. All-frames MSE $\times 10^{-3}$.

	Reference, 3 seeds			back001, 3 seeds			All 6 checkpoints		
Members	ext	int	long	ext	int	long	ext	int	long
single decode	1.65	3.08	15.28	1.77	2.77	14.15	1.65	2.84	14.40
mixture (current protocol)	1.58	2.93	12.67	1.69	2.67	12.95	1.59	2.73	12.68
mixture + scalar shrink to climatology	1.58	2.92	11.97	1.69	2.66	12.04	1.59	2.73	11.97
mixture + Wiener to climatology	1.58	2.91	12.01	1.69	2.65	12.03	1.59	2.72	11.97

Table 6: LSiM $\times 10^{-1}$ (lower is better) for the shortlisted variants: mean over the 3 seeds, and of the 3-seed average.

		ext		int		long	
Model	Variant	seeds	average	seeds	average	seeds	average
Reference	mixture	1.72	1.75	1.87	1.86	5.74	6.07
	+ shrink to clim.	1.78	1.81	1.91	1.91	7.00	7.10
	+ Wiener to clim.	1.80	1.82	1.91	1.91	6.85	6.97
back001	mixture	1.81	1.85	1.83	1.82	6.09	6.50
	+ shrink to clim.	1.86	1.90	1.87	1.85	7.23	7.35
	+ Wiener to clim.	1.89	1.93	1.86	1.85	7.04	7.20
Single decode, one seed (for reference)		1.41	–	1.71	–	4.06	–
ACDM / U-Net$_{ut}$		1.29 / 1.09		1.38 / 0.95		3.83 / 3.81	

What works.

- **Seed averaging is the largest gain.** The 3-seed average of the mixture forecasts lowers ext by 0.19 (both models), int by 0.22–0.23 and long by 0.36 (reference) and 0.50 (back001) against the single-seed mean. The reference's 3-seed average (1.58) is ahead of U-Net$_{ut}$ on ext (1.60); back001's (2.67) is ahead of ACDM on int (2.92). Averaging all six checkpoints is no better than the better 3-seed average on any case.
- **Climatology shrinkage helps long only**: -0.90 ± 0.05 (reference) and -1.23 ± 0.05 (back001), paired; on the 3-seed averages -0.70 and -0.91. The calibrated gains stay close to 1 up to lead 58, so ext and int barely move (≤ 0.03). It helps back001 more, which nearly closes back001's long deficit (12.22 vs 12.13, single-seed means).
- **Scale dependence adds nothing**: the per-wavenumber Wiener filter and one scalar per lead and channel are within 0.03 of each other; the scalar is the simpler choice.

What does not.

- **A lead-dependent mixture width** is no better (reference long $+0.11 \pm 0.03$): the train-optimal width is 0.02 for the first leads and close to the calibrated one after lead 30, and the first leads carry little error.
- **Shrinking towards the forecast's own mean** gives 30–60% of the climatology gain (-0.26 to -0.72 on long). Applied to the single decode instead of the mixture, it is worse on ext and int ($+0.02$ to $+0.03$ against the mixture) and only matches it on long for the reference.

Costs. All of it lowers LSiM quality: shrinkage adds 0.95–1.26 ($\times 10^{-1}$) on long and 0.03–0.08 on ext and int, and seed averaging adds a little more on ext and long, because both reduce the fluctuation energy of uncertain forecasts. Seed averaging triples inference. Neither changes the data assimilation, which uses the latent model of one checkpoint; DA with a seed ensemble (three independent 4D-Var analyses, averaged) has not been run.

Post-processing to report. Primary: 3-seed average + mixture + scalar climatology shrinkage, calibrated on training rollouts. Report single-model numbers and LSiM alongside it.

7 Findings from the investigation

- **The wake is a noisy oscillator.** Phase speed varies by about $\pm 14\%$ within a simulation and is unpredictable from the window ($R^2 \approx 0.19$); ω(Ma) is jagged; about 29 oscillator pairs within $\pm 20\%$ of the shedding frequency carry 66% of the latent variance.
- **Long-lead hedging is dephasing, not damping**, and averaging decodes in pixel space, $\mathbb{E}[D(z)]$, beats shrinking the latent, $D(\mathbb{E}[z])$. This is the basis of the mixture readout.
- **The short-lead floor is a generalisation gap**, not a capacity limit: reconstruction error 0.004–0.07 on training simulations against about 0.35 on Tra_int; a 256-mode POD reaches 0.20.
- **DA error is identifiability and overfitting of model error.** An oracle test shows the model's trajectory family can fit both t_0 and the future; plain 4D-Var instead fits the observations with a wrong t_0. Removing damping at DA time (0.0958 vs 0.0963) or warping the phase speed (0.0961 / 0.0957) does not help. The backward term (tested at weight 0.1) cuts the oracle future misfit from 0.040 to 0.015, the likely mechanism behind back001's DA gain.

8 Open items

- DA with a seed ensemble (average of three 4D-Var analyses), to see whether the ensemble gain carries over to DA.
- back001 combined with the conditional fixed point μ(Ma) (both help int; neither hurts DA).
- Next candidates, not yet run: a POD-backbone decoder (attacks the generalisation gap); leave-block-out Mach conditioning; training with the mixture readout inside the loss; test-time refinement of z_0 on the input frames; a tied quadrature multiplet.
- Paper: switch the main tables to the chosen CT-KAE and its DA results (`da_v3`).